

MURALI SAI BUDDAKKAGARI VENKATA

+1 (201)-275-6592 | murali140824@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Applied AI Engineer with production experience in LLM fine-tuning, agentic system design, and RAG pipeline deployment. Fine-tuned open-source models (Llama, Mistral) using LoRA/QLoRA; built multi-agent orchestration systems (LangGraph, GPT-4o) achieving ~85% autonomous task completion. Designed RAGAS-based eval frameworks tracking faithfulness, relevancy, and recall. Researching domain-adaptive ensemble learning for medical nucleus detection at Stevens Institute of Technology.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C++, PySpark, TypeScript
- **AI & Machine Learning:** Prompt Engineering & Design, Agent Evaluation (RAGAS), Retrieval-Augmented Generation, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, CrewAI, HuggingFace, Transformers, PyTorch, TensorFlow, FastAPI, ChromaDB, Scikit-learn, Embedding Models, LLMops, LangSmith, Fine-tuning, LoRA / QLoRA / PEFT, vLLM, Gemini
- **Infrastructure & DevOps:** AWS (ECS Fargate, S3), Google Cloud Platform, BigQuery, Azure, Docker, PostgreSQL, GitHub Actions, CI/CD, OpenAI API, Claude API, Pinecone, MLflow

EXPERIENCE

Querkey Inc

Jun 2025 - Aug 2025

Data Science Intern

CA, US

- Accelerated ML pipeline throughput by 40%, processing 200+ MP4 files daily, by engineering Python/PySpark ingestion pipelines on GCP/BigQuery that eliminated manual validation bottlenecks across 3 production data streams.
- Cut production ML workflow failure rate by 60% across 100K+ records by deploying a Dockerized MLOps observability stack (MLflow, HuggingFace model monitoring, ChromaDB, FastAPI), enabling real-time anomaly detection before downstream impact.

VIEW Synergy

Jan 2024 - Apr 2024

AI and Automation Intern

KA, IND

- Reduced manual task overhead by 35% for 100+ employees by architecting LLM-powered automation pipelines (LangChain, LlamaIndex, Azure OpenAI, TypeScript) for intelligent document processing and workflow orchestration.
- Achieved sub-200ms API latency in production by shipping RAG pipelines (OpenAI API, Docker, PostgreSQL) with agentic tool-use and retrieval-augmented responses in a compliance-sensitive environment.

PROJECTS

Agent Orchestration System with Tool Use, Memory, and Human in the Loop

- Built a production multi-agent system (LangGraph, GPT-4o, Claude Haiku) with 3-layer agent hierarchy, parallel task execution, and episodic + semantic two-tier memory (Redis + ChromaDB), achieving ~85% autonomous task completion across long-horizon workflows.
- Reduced human intervention to 15% of edge cases by designing a Human-in-the-Loop escalation system with 5 automated triggers and 4 approval levels, with full audit trail validated across fintech and legal compliance requirements.

SEC EDGAR RAG Filing Analyzer

- Improved RAG eval scores (faithfulness 0.57→0.65, relevancy 0.58→0.68, recall 0.57→0.70) across SEC 10-K filings by iteratively tuning retrieval strategies, chunk sizing, and context construction using LangChain, LlamaIndex, and ChromaDB, with evals in CI via RAGAS, cutting manual eval effort by ~70%.
- Deployed a compliance-grade RAG system with RBAC, MNPI guardrails, and SEC 17a-4/FINRA 4511-aligned audit trail on AWS ECS Fargate (Terraform + GitHub Actions CI/CD) with sub-second retrieval against SEC filing corpora.

WorkFit: Multimodal Posture Monitoring System

- Built a real-time multimodal posture classification system (MobileNetV2, Random Forest, MediaPipe) fusing computer vision and flex sensor streams via majority-vote ensemble, achieving 84.8% CNN accuracy and 81.48% sensor model accuracy across an 80,000-image augmented training dataset.
- Awarded Best Research Paper at ICICC 2024 for designing a 3-modality MQTT pipeline integrating webcam-based image classification, flex sensor data (ESP8266/Arduino), and MediaPipe landmark scoring with real-time notifications and MongoDB persistence.

EDUCATION

Stevens Institute of Technology

Master of Science, Applied Artificial Intelligence

May 2026

NJ, US

PES University

Bachelor of Technology, Computer Science (Specialization in ML)

May 2024

KA, IND